

HDOC DAY-31

Title: Menu-Driven 2D Array / Matrix Operations in C

1. Problem Statement

Write a C program to perform the following menu-driven operations on a given 2D array:

1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit.

2. Algorithms

Algorithm 1: Matrix Sum Calculation

1. Input dimensions R (rows) and C (columns).
2. Input matrix elements into a 2D array `matrix[R][C]`.
3. Initialize variable `total_sum = 0`.
4. Iterate i from 0 to $R - 1$ and j from 0 to $C - 1$:
 - `total_sum = total_sum + matrix[i][j]`
5. Print `total_sum`.

Algorithm 2: Row Sum and Array Storage

1. Declare 1D array row_sums[R].
2. Iterate outer loop i from 0 to $R - 1$:
 - Set row_sums[i] = 0.
 - Inner loop j from 0 to $C - 1$: row_sums[i] += matrix[i][j]
 - Output sum of row $i + 1$.
3. Output the combined row_sums array.

Algorithm 3: Matrix Transpose

1. Create a 2D array transpose[C][R].
2. Iterate i from 0 to $R - 1$ and j from 0 to $C - 1$:
 - Set transpose[j][i] = matrix[i][j]
3. Display the transposed matrix of dimension $C \times R$.

3. Test Case Table

Test Case	Option	Matrix Input	Expected Output	Actual Output	Status
1	Total Sum	2×3 , [[1, 2, 3], [4, 5, 6]]	Sum of all elements = 21	Sum of all elements = 21	Pass

Test Case	Option	Matrix Input	Expected Output	Actual Output	Status
2	Row Sums	2×3 , [[1, 2, 3], [4, 5, 6]]	Row 1: 6, Row 2: 15		

Array: [6, 15] | Row 1: 6, Row 2: 15

Array: [6, 15] | Pass | | **3** | Transpose | 2×3 , [[1, 2, 3], [4, 5, 6]] | 3×2 Matrix:

14

25

3 6 | Matched 3×2 matrix output | Pass | | **4** | Exit | Choice
4 | Exiting program... | Program terminated cleanly | Pass |

4. Source Code Listing

```
#include <stdio.h>

int main(void) {
    int r, c;
    printf("Enter the number of rows and columns: ");
    scanf("%d %d", &r, &c);

    int matrix[r][c];
    printf("Enter the matrix elements:\n");
    for (int i = 0; i < r; i++) {
        for (int j = 0; j < c; j++) {
            scanf("%d", &matrix[i][j]);
        }
    }

    int choice;
    do {
        printf("\nEnter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: {
                int total_sum = 0;
                for (int i = 0; i < r; i++)
                    for (int j = 0; j < c; j++)
                        total_sum += matrix[i][j];
                printf("Sum of all elements = %d\n", total_sum);
                break;
            }
            case 2: {
                int row_sums[r];
                printf("Row sums:\n");
                for (int i = 0; i < r; i++) {
                    row_sums[i] = 0;
                    for (int j = 0; j < c; j++) row_sums[i] += matrix[i][j];
                }
            }
        }
    } while (choice != 0);
}
```

```

        printf("Sum of row %d = %d\n", i + 1, row_sums[i]);
    }
    printf("The row-sum array contains: [");
    for (int i = 0; i < r; i++) {
        printf("%d%s", row_sums[i], (i < r - 1) ? ", " : "");
    }
    printf("]\n");
    break;
}
case 3: {
    int transpose[c][r];
    for (int i = 0; i < r; i++)
        for (int j = 0; j < c; j++)
            transpose[j][i] = matrix[i][j];

    printf("Transpose of the matrix:\n");
    for (int i = 0; i < c; i++) {
        for (int j = 0; j < r; j++) printf("%d\t", transpose[i][j]);
        printf("\n");
    }
    break;
}
case 4: printf("Exiting...\n"); break;
default: printf("Invalid choice!\n"); break;
}
} while (choice != 4);

return 0;
}

```

4. Execution & Verification Screenshots

```

kali@kali:~$ cd /root/.vscode/extensions
kali@kali:~/vscode/extensions$ cd HDOC_Day31
kali@kali:~/HDOC_Day31$ ls
main day31.c
kali@kali:~/HDOC_Day31$ gcc day31.c -o day31
kali@kali:~/HDOC_Day31$ ./day31
Enter the number of rows and columns: 2 3
Enter the matrix elements:
1 2 3
4 5 6

Menu:
1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit
Enter your choice: 1
Sum of all elements = 21

Menu:
1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit
Enter your choice: 2
Row sums:
Sum of row 1 = 6
Sum of row 2 = 15
The row-sum array contains: [6, 15]

Menu:
1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit
Enter your choice: 3
Transpose of the matrix:
1      4
2      5
3      6

Menu:
1. Find the sum of all elements in a matrix.
2. Find the sum of each row of a matrix and store it in an array.
3. Find the transpose of a matrix.
4. Exit
Enter your choice: 4
Exiting program...
kali@kali:~/HDOC_Day31$

```